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Towards evidence-informed rare disease surveillance and policy: First insights from Belgium's national registry (2015–2025)

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The Central Registry of Rare Diseases (CRRD) underpins Belgium's national surveillance infrastructure for rare diseases.
This study characterises the registered population between 2015 and 2025, describing diagnostic pathways, referral patterns and coding practices.
The registry generates evidence to support epidemiological surveillance, research, health system organisation and policy.

WHY IT MATTERS

Structured real-world rare disease data is scarce, heterogeneous and fragmented

Lack of harmonised data limits surveillance and policy development

National registries are essential for evidence-informed care planning

WHAT IS THE CRRD

- National population-based registry
- Data collected at diagnosis only
- Cross-sectional design
- Pseudonymised minimal dataset

INCLUSION CRITERIA

- Patients with confirmed rare disease diagnoses
- Diagnosed at accredited genetic centres in Belgium

STANDARDS & INTEROPERABILITY

- ORPHAcodes (Orphanet nomenclature)
- EU RD Platform Common Data Elements

LIMITATIONS & BARRIERS

- Incomplete national coverage
- Heterogeneous coding practices
- Missing or inconsistent data
- Limited interoperability across EHR systems
- Cross-sectional design without longitudinal follow-up

KEY FINDINGS

DISEASE CODING PRACTICES

While ORPHAcodes were the most frequently used coding system within the registry (70.2%, representing 1,423 distinct disease codes), the coexistence of multiple coding systems reinforces the need for national harmonisation and improved interoperability.

9,551 Registered unique patients

70.2% Registrations with ORPHAcodes

1,423 Distinct rare diseases

DIAGNOSTIC DELAY

Across all analysed cases (N = 9,551 from 7 centres), the median diagnostic delay was approximately 1.5–1.7 years, although delays ranged from very short to more than a decade in some patients.

1.5 – 1.7 years

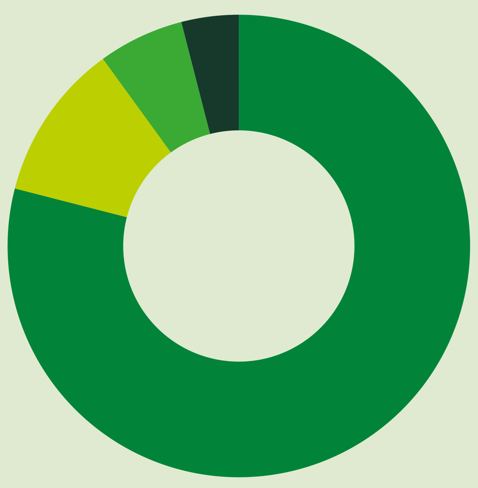
Median diagnostic delay

DISEASE ONSET

The median age at symptom onset was 3 years, indicating that most rare diseases manifest in childhood, although onset ranged from the neonatal period to adulthood, reflecting their diverse natural histories.

	Neonatal (onset = 0 y)	Infantile (0 < onset ≤ 2 y)	Childhood (2 < onset < 18 y)	Adult (onset ≥ 18 y)
♀	31.7%	13.7%	19.9%	34.7%
♂	35.8%	18.1%	18.0%	28.2%

REFERRAL PATTERNS



79.4% Medical specialists

10.9% Patient-initiated

5.6% General practitioners

4.1% Other

INTERPRETATION

First structured overview of rare disease surveillance in Belgium

Reveals diagnostic and referral patterns

Heterogeneity in rare disease documentation underscores its intrinsic complexity

Demonstrates the importance of interoperable EHR infrastructure

FUTURE PERSPECTIVES

Mandatory registration under the second Belgian Plan for Rare Diseases (2026–2030)

Expansion to additional data providers as of 2025

Data linkage with disease-specific registries and Belgian National Registry

Enhance representativeness, harmonisation and interoperability

Strengthen the CRRD's role as a digital infrastructure for evidence-informed care and policy

MORE ABOUT THE CRRD

Learn more about the CRRD and access related reports and publications

The CRRD lays the foundation of Belgium's emerging national surveillance system for rare diseases. Strengthening coverage, harmonisation, and interoperability will support future reporting, continuous improvement, and the development of a more coordinated and patient-centred rare disease ecosystem.

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